

Response to the independent SC14 MSE desk review

Overall response

The review concludes that the report is a strong scientific summary of a credible MSE and supports its use to continue SC14 discussion and identify a provisional shortlist. Its central concern is consistency across the report, CMP specifications, registries, source and public data, figures, Slick results, and annual-application material.

The authors accept that central finding. The full MSE report, SC14 questions and tasks paper, technical paper, and this response use the same final results. The immediate request is for SC14 to agree that the scientific work is sound, agree the decision criteria, identify a provisional short list, and provide management guidance. Adoption can follow completion of the instructions and checks needed for annual use.

Response to the Terms of Reference

Table 1: Response to the six review questions.

ToR finding	Response
Design is broadly consistent with SCW17, but reported results and public files are not consistently aligned	Save one final SC14 set of results and use it to remake all public products; document how the candidates changed from SCW17 to the final comparison
Overall MSE structure is appropriate, but tuning, allocation, precision, performance measures, and criteria for alternative cases need further support	Put definitions and complete CMP-by-OM results in the technical paper and ask SC14 to agree the minimum biological performance requirements and how alternative cases should be used. Treat allocation analysis as complete; add the author's explanation of why the existing work is sufficient
Uncertainties and limitations are clear in prose but not always supported by one checkable numerical account	Add one biological-performance table, results for each component in the two-stock OMs, uncertainty due to the number of runs, effort and low-catch results, and a comparison using alternative reference points
A provisional shortlist may be supportable, but the supporting results and annual rules are incomplete	Ask for a provisional shortlist and a list of work to finish rather than immediate adoption
Material inconsistencies occur in settings, data timing, candidate membership, results, and figure construction	Require the important numbers to agree across all products before publication and identify every product that is no longer current
Scientific findings and management choices are not always separated	Separate minimum biological performance, results under alternative assumptions, and management preferences, and place management questions in the decision table

Response to the main issues

Table 2: Response to the review's eight principal issues.

Priority and review issue	Response	Work needed	Timing
High — align published files	Agreed	One named final version; exact code revision; list of simulation runs and files; remade report, CMP list, CMP and annual-use pages, figures, CSVs, and Slick file; automatic report showing that the important numbers agree	Before SC14 submission
High — limit current request to shortlist consideration	Agreed	Revised requested-advice text and explicit pre-adoption completion list	Before SC14 submission
High — clarify minimum biological requirements and difficult cases	Agreed	Explain the relationship between COMM8 and K60; agree measures of the chance of falling below a biomass limit; provide a CMP-by-OM table including om23; show results for each stock component; explain how plausible each OM is and how it should affect the decision	Requirements requested at SC14; completed before adoption

Priority and review issue	Response	Work needed	Timing
High — describe HS versus PR as complete configurations	Agreed	Revised terminology; matched illustration used only to explain rule shape	Before SC14 submission
Medium — assess whether 500 simulations are sufficiently precise	Agreed	Description of how uncertainty was represented, use of the same random draws, uncertainty ranges, and whether conclusions change with more runs or different draws	Before final recommendation
Medium — treat scorecard rankings as preliminary	Agreed	First determine which CMPs meet the minimum biological performance requirements; retain the original results; show how rankings change with measures, weights, candidates, periods, and OMs	Before SC14 submission for framing; complete before final recommendation
Medium — document and test index errors	Agreed in stages	Final calculations and values; record of overlapping years and data changes; checks against observed indices; tests of gradual change, abrupt change, missing surveys, regional balance, and index reliability	Core documentation before SC14; additional tests in continuing work
Medium — show candidate evolution and retain reviewer caveats	Agreed	SCW17-to-current comparison table; dated decision records; reviewer comments separated from author response	Before SC14 submission

How to tell when the work is complete

Agreement among the published files

Completion requires more than checking that a file exists or has the expected size. Automatic checks compare CMP settings, candidate membership, tuning, performance statistics, time periods, labels, two-stock components, scorecard inputs, public downloads, and reported statements. A checksum identifies a file; it does not establish scientific consistency among files.

Minimum biological performance requirements

The 60% dynamic Kobe-green probability remains a common tuning target, not the only requirement for biological performance. The response should explain how this target relates to the COMM8 objectives and should report the chance of low biomass and overfishing. Results must be shown for each relevant stock component and difficult OM. A decision to accept a known weakness must be identified as a management judgment.

Ready for annual use

Completion requires a complete set of instructions from which two independent analysts obtain identical advice. It must address units, providers, cut-off dates, missing and revised data, weighting, prior advice, annual limits, rounding, allocation, realized catch, effort, fallback, verification, and ECP rules for who reviews a problem and what action can be taken.

Spatial results and the allocation comment

The allocation analysis is complete for this paper. The author's response will explain the purpose of the fixed-split tests, how they relate to the allocation method in the CMP, and why no additional allocation analysis is needed. The spatial results remain relevant where they help SC14 understand stock structure, movement, or regional monitoring.

Banking and borrowing

The updated runs may be reported as work in progress. A conclusion about the incremental effect of banking and borrowing requires a matched without-banking-and-borrowing comparison using identical OM draws and all other CMP settings held constant.

Overall conclusion

The authors thank the reviewer and accept the central recommendation that the scientific synthesis and its supporting products must present one consistent account. The MSE is sufficiently advanced for Scientific Committee review and provisional shortlisting. Final recommendation or adoption should depend on completion of the checks on published results, minimum biological performance requirements, instructions for annual use, and exceptional-circumstances items recorded in this response.

The complete reviewer text will accompany this response or be submitted as a separate official information paper. These response tables do not replace the independent review.